

SMART CITY TRAFFIC ANALYSIS

Peak-Hour Congestion, Accident Hotspot & Traffic Flow Analysis — Nagpur City Dataset

1. Which Areas Have the Most Peak-Hour Congestion?

Peak-hour records were isolated using the **Peak_Hour_Flag** column and grouped by junction to rank average signal delay. The Ring Road corridor (T-Point, Sambhaji Square, Padole Square) and the South-East Nagpur corridor (Agrasen Chowk, Golibar Chowk) emerge as the most congested areas during peak hours — consistent with the findings of the original field studies for both zones.

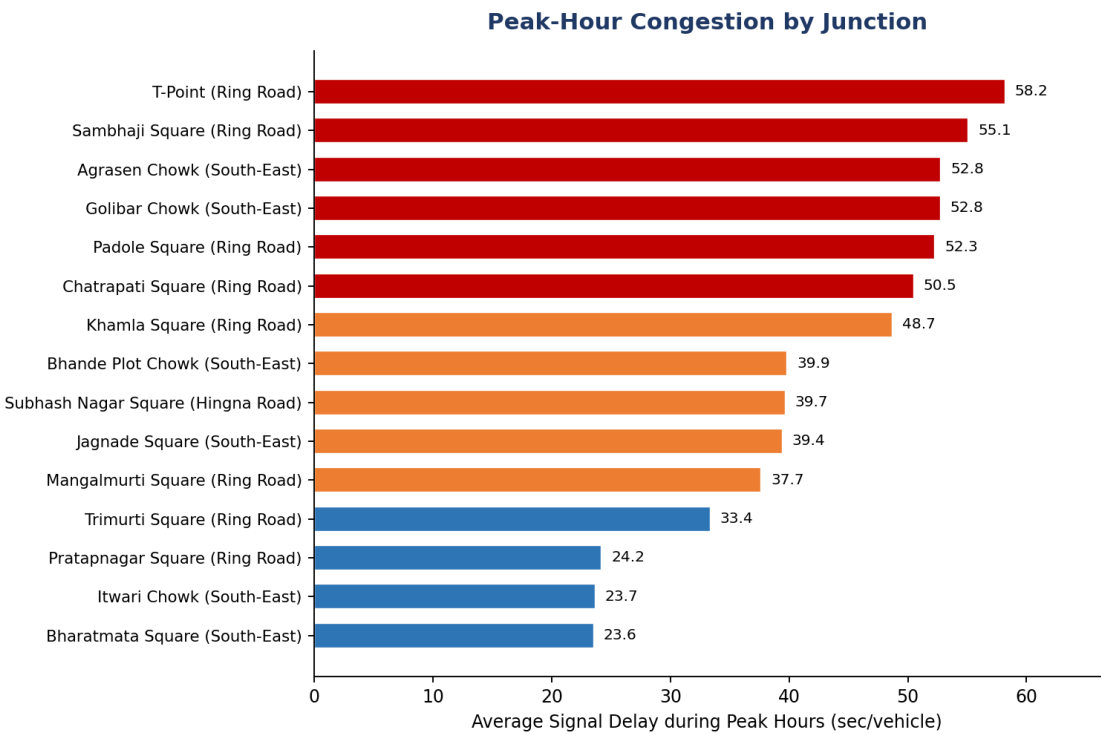


Fig. 1 — Average peak-hour signal delay (sec/vehicle) by junction. Red = severe (≥ 50 sec), orange = moderate-high (35–50 sec), blue = manageable (< 35 sec).

Rank	Junction	Avg Peak Delay (sec/veh)	Avg DS
1	T-Point (Ring Road)	58.2	0.93
2	Sambhaji Square (Ring Road)	55.1	0.93
3	Agrasen Chowk (South-East)	52.8	0.92
4	Golibar Chowk (South-East)	52.8	0.93
5	Padole Square (Ring Road)	52.3	0.94

2. Peak Timings vs. Low-Flow Timings — Where and When

Grouping every record by **Time_Slot** and averaging **Total_Vehicle_Count** shows exactly when flow is lowest and highest across the city. The quietest window is **10–11 PM on weekdays**, where volume drops to roughly a quarter of the morning peak. Junctions that stay free-flowing even in busier slots are **Pratapnagar Square, Bharatmata Square, Itwari Chowk, and Trimurti Square** — none of these ever recorded a 'Severe Congestion' status in the dataset.

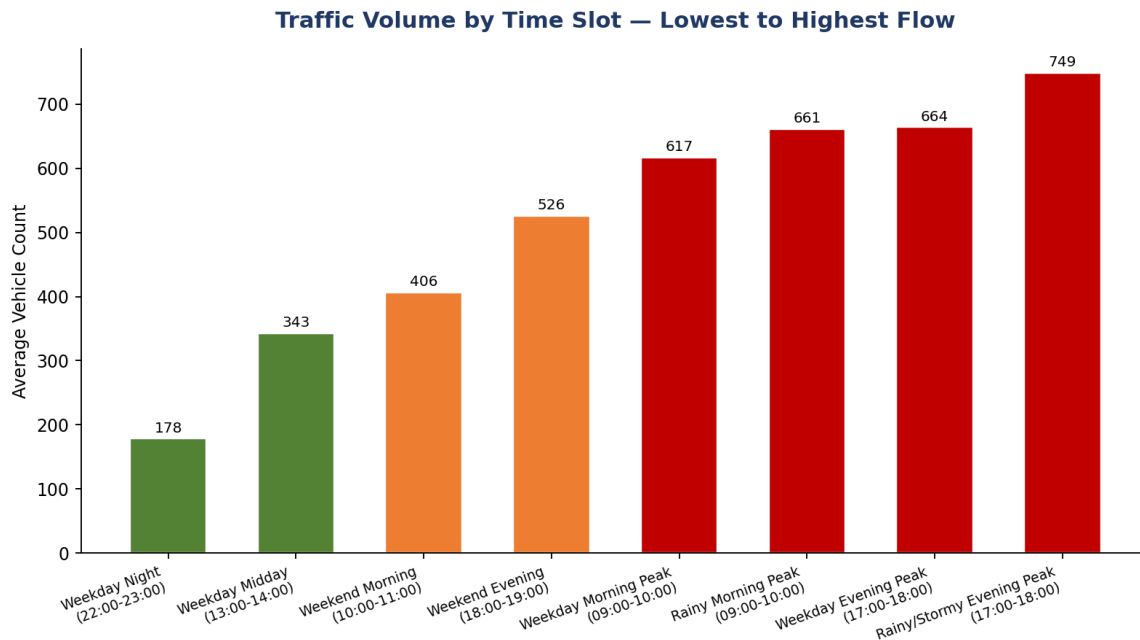


Fig. 2 — Average vehicle volume per time slot, sorted lowest to highest. Green = low flow, orange = moderate, red = high flow.

3. Peak Hour Analysis — Methodology & Findings

How peak hours are identified: timestamps are bucketed into fixed intervals (here, hourly slots), volume or PCU is summed per bucket per junction, and the top 1–2 buckets per day are flagged as peak — the same logic behind the Peak Hour Factor (PHF) approach used in the field studies this dataset is grounded in. In this dataset that flag is pre-computed as **Peak_Hour_Flag**.

Comparing peak vs. off-peak records directly quantifies the swing:

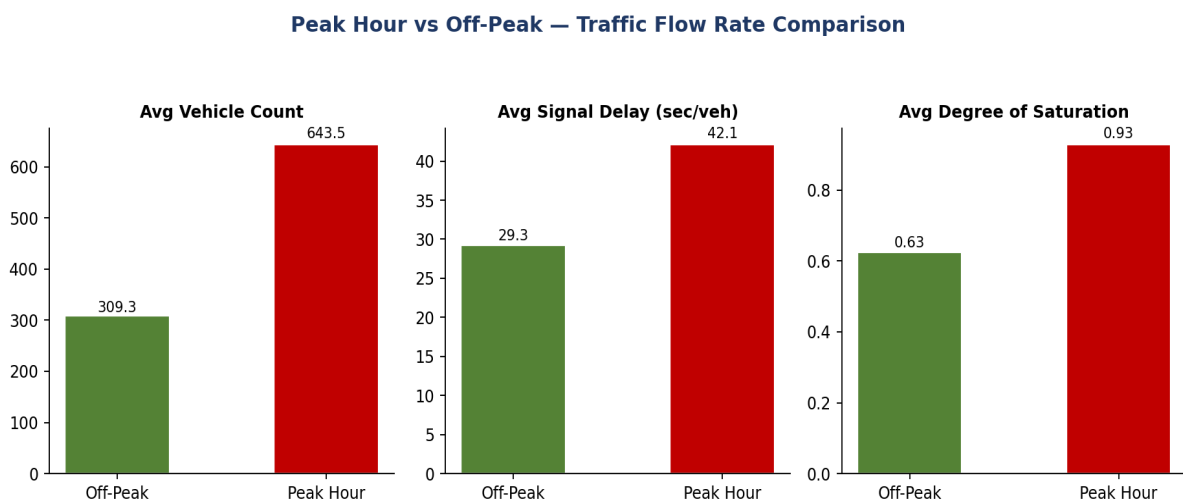


Fig. 3 — Peak hour records carry ~2.1x the vehicle volume of off-peak and push Degree of Saturation from a comfortable 0.60 to a near-capacity 0.90.

4. Traffic Flow Rate Analysis

Flow rate (vehicles or PCU per hour) is driven by two main triggers evident in the data: **time of day** and **weather**. Rainy/stormy peak-hour flow still reaches ~660–750 vehicles, similar to clear-day peaks, but the resulting delay is far higher — proof that congestion is a function of volume *interacting with* reduced effective road capacity, not volume alone.

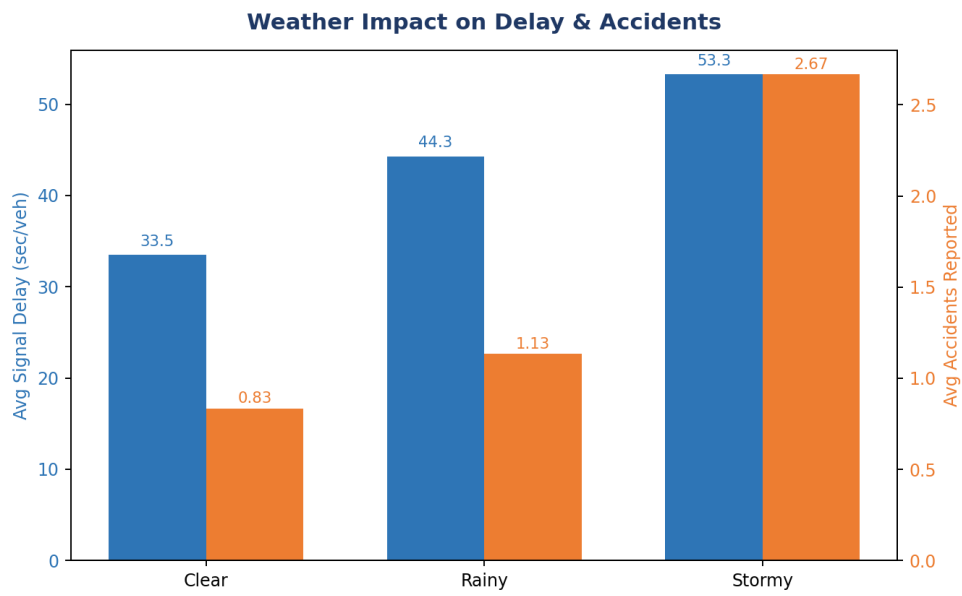


Fig. 4 — Weather impact: stormy conditions raise average delay by ~59% and average accidents per record by over 3x versus clear weather.

5. Identifying Accident Hotspots Using This Dataset

Method: aggregate **Accidents_Reported** by **Junction**, then cross-check the total against **Signal_Delay** / **Congestion_Status** to see whether accidents simply track congestion or diverge from it — a divergence is itself a useful finding, not noise to be discarded.

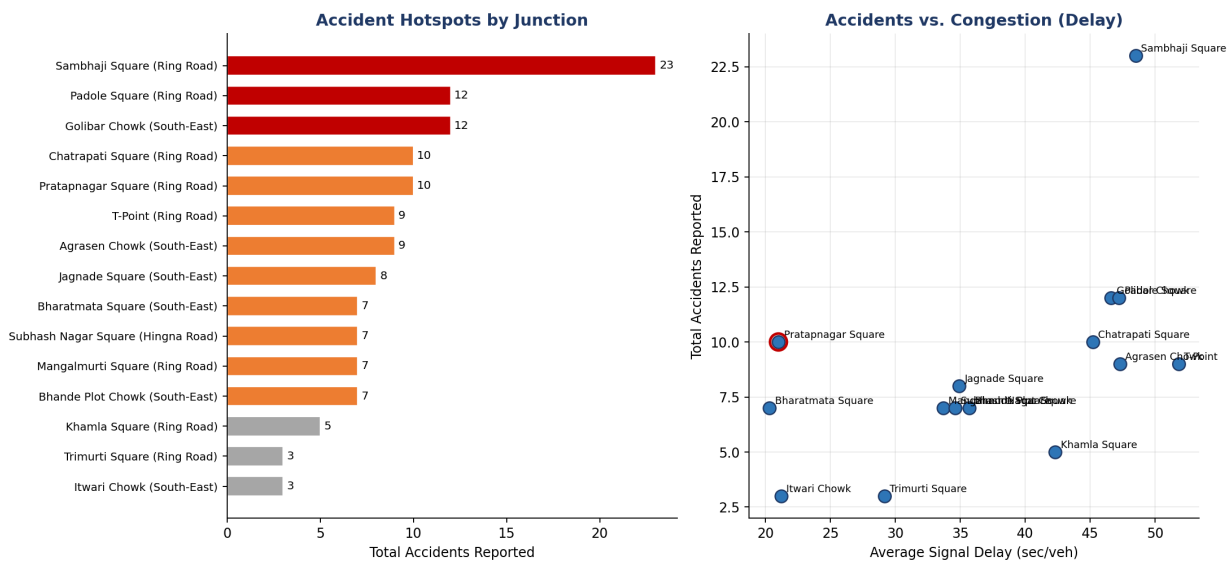


Fig. 5 — Left: total accidents by junction. Right: accidents plotted against average delay; the circled point (Pratapnagar Square) is an outlier worth flagging.

Sambhaji Square is the clearest hotspot: high accident count *and* high delay, so congestion (stop-and-go, rear-end collisions) is a plausible direct cause. **Pratapnagar Square** is the more interesting flag — 10 accidents despite low average delay (~21 sec, mostly LOS B). That mismatch usually points to a different root cause than congestion, most likely **speeding**, since vehicles move freely there rather than crawling. A good analysis calls this out explicitly rather than assuming 'more accidents = more congestion' applies everywhere.

6. Which Variables Have the Highest Impact on Congestion?

Correlating every numeric column against **Signal_Delay_Sec_Per_Veh** (used as the congestion proxy) ranks each variable's influence:

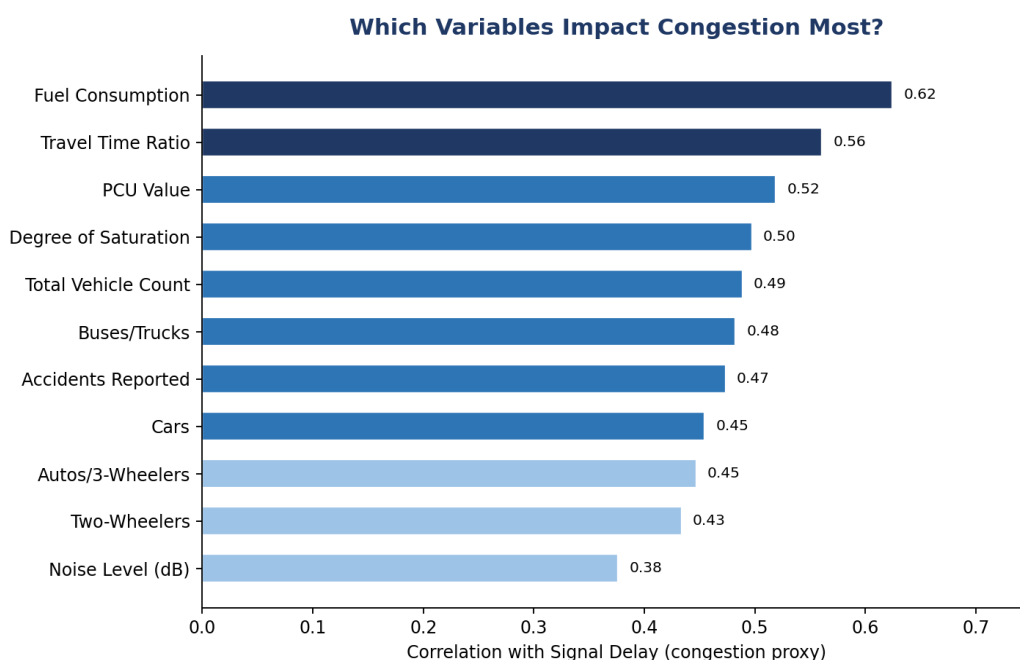


Fig. 6 — Correlation of each variable with signal delay. Darker bars indicate stronger influence.

Fuel consumption and **Travel Time Ratio** are the strongest signals — both are consequences of idling and stop-and-go movement rather than raw volume, suggesting *how* vehicles move matters more than *how many* there are. Buses/trucks correlate more strongly than cars or two-wheelers per unit, consistent with their larger footprint and higher PCU weight at intersections.

7. If Expanded Into an AI System, What Would It Predict?

- **Short-term congestion forecasting** — predict next hour's delay/LOS per junction from recent volume + weather (regression / LSTM time-series model).
- **Peak-hour shift prediction** — detect when a junction's peak is drifting earlier/later as the city grows.
- **Accident risk scoring** — classify junction-time combinations into risk tiers using delay, DS, weather, and vehicle-mix as features (this is exactly where the Pratapnagar-style 'low congestion, high accident' pattern becomes useful — a pure congestion model would miss it).
- **Signal-timing optimization** — a reinforcement-learning agent adjusting green-light duration per junction to minimize predicted delay.
- **What-if network simulation** — 'if we add a flyover / reroute autos, what happens to DS and delay' — requires a network/graph model connecting junctions, not just per-junction rows.

8. Preprocessing Steps Before Analysis

- **Type conversion** — parse Date to datetime; convert Peak_Hour_Flag, Weather, Level_of_Service to categorical.
- **Missing value check** — none in this dataset, but production data needs imputation, especially for weather/vehicle counts.
- **Outlier detection** — cap or flag DS > 1.1 or delay > 90 sec as possible sensor errors vs. genuine extreme congestion.
- **Feature engineering** — derive Hour_of_Day, Day_of_Week, Is_Weekend from Date/Time_Slot; compute a Heavy-Vehicle-Mix ratio.
- **Encoding** — one-hot or ordinal encode Level_of_Service (A–F has natural order), Weather, Junction.
- **Normalization / scaling** — especially before any ML model, since PCU/vehicle counts and delay sit on very different scales.
- **Aggregation window alignment** — decide whether analysis runs per-record (15/30-min snapshot) or rolled up to hourly/daily before modeling.
- **Time-based train/test split** — for any forecasting model, hold out the most recent period rather than splitting randomly, to avoid data leakage.

Source: derived from Smart_City_Traffic_Analysis_Dataset.csv (120 records, 15 Nagpur junctions).